SLIIT FACULTY OF COMPUTING · CDAP R26-IT-162

FloodSense AI

Intelligent IoT Microclimate Sensing & Dual-Stage Predictive Flood Inundation Modeling

Hyperlocal rainfall forecasting and distance-based dynamic risk scoring for Kelani River Basin communities.

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PROBLEM CONTEXT · KELANI RIVER BASIN

The warning arrives too late, and says too little

Across Colombo, Gampaha, and Kegalle, recurring monsoonal inundation places 500,000+ vulnerable citizens at risk.

LKR 50B+	annual economic loss
300,000+	people displaced annually
6–12h	current warning gap



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REVIEWER FEEDBACK · SYSTEM EVOLUTION

Panel comments moved the prototype toward a deployable warning system

Four upgrades changed what the system can sense, predict, and communicate.

01 DETECTION

Single sensor → multi-factor IoT

Calibrated tipping bucket; reed-switch debouncing ISR; 14-variable Open-Meteo fusion.

02 DATA PIPELINE

Manual logs → MQTT streaming

Sub-second HiveMQ telemetry; local non-volatile buffering; automatic reconnect watchdog.

03 PREDICTION

Binary alert → dynamic risk

Continuous 0–100% risk; arrival time in hours; water height with cm precision.

04 BENEFICIARIES

Generic message → DFRS

Distance-based personalization; automated Sinhala, Tamil, and English notifications.

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SYSTEM ARCHITECTURE · DATA FLOW

One pipeline connects a rainfall pulse to an inundation decision

Component A supplies live microclimate data and rainfall forecasts. Component B converts them into localized risk and action.

01

Sense

ESP32, tipping bucket, temperature, humidity, and JSON telemetry.

02

Fuse

Open-Meteo atmospheric variables plus antecedent rainfall buffers.

03

Predict

24h, 48h, and 72h rainfall depths feed the two-stage flood model.

04

Decide

Occurrence, risk score, arrival time, depth, and trilingual alerts.

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COMPONENT A · EMBEDDED IOT

Reliable edge telemetry begins with a calibrated rainfall pulse

Measure

ESP32 dual-core Xtensa LX6, 240 MHz; dual-chamber tipping bucket; 0.20 mm per tip; DHT22 / BME280 sensing; 3.7V 18650 battery; 6V 5W solar panel.

Clean

GPIO 4 falling-edge ISR; 100 ms software debounce; 5-sample moving median filter; dynamic intensity calculation; watchdog recovery.

Transmit

MQTT v3.1.1 through HiveMQ Cloud; serialized JSON payloads; SPIFFS fallback; typically <280 ms edge-to-cloud ingestion.

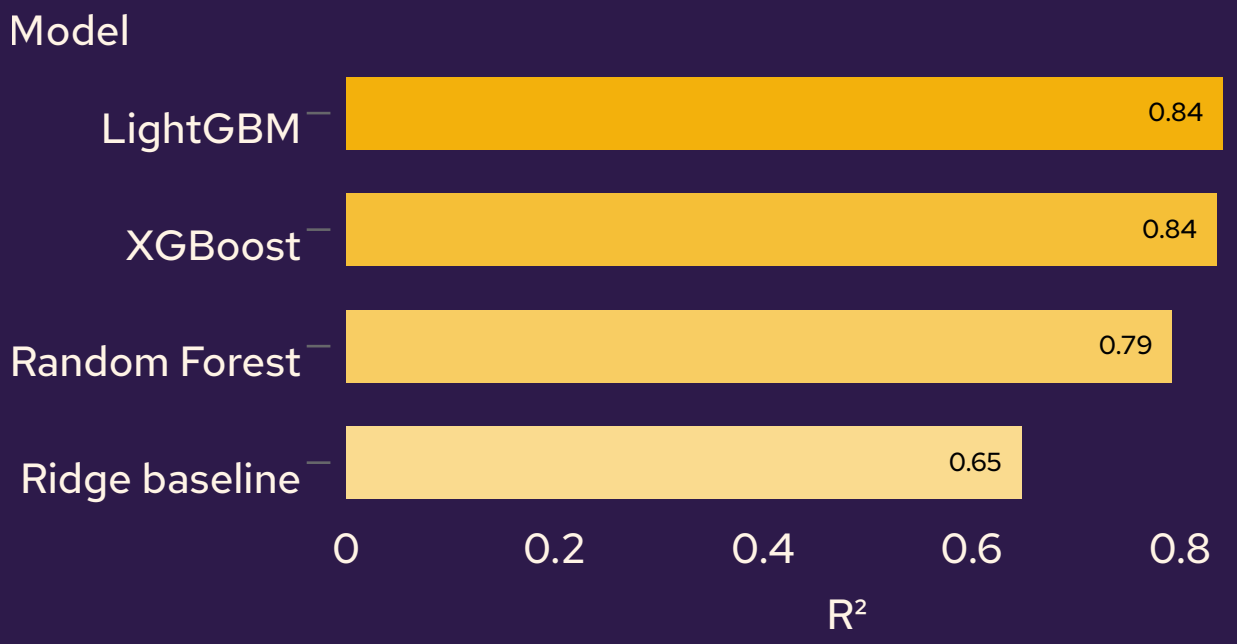


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COMPONENT A · RAINFALL FORECASTING

Atmospheric fusion extends local rainfall into three horizons

Day 1 / 24h benchmark: LightGBM leads the comparative evaluation.



24h · 48h · 72h

Forecast horizons

Fourteen atmospheric variables, rolling rainfall windows, API with k = 0.88, wind-vector decomposition, and cyclical time encoding.

LightGBM: MAE 2.84 mm · RMSE 4.12 mm · 30 ms inference latency

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COMPONENT B · TWO-STAGE FLOOD ML

Flood risk becomes actionable when occurrence, severity, and timing are separated

01

STAGE 1A

Will flooding occur?

XGBoost classifier estimates calibrated P(Flood). F1-score: 0.941.

02

STAGE 1B

How severe is the risk?

XGBoost regressor produces a 0–100 score. RMSE: 3.12.

03

STAGE 2

When and how deep?

Multi-output XGBoost predicts arrival time in hours and peak water depth in meters.

≥70	50–69	30–49	<30
Very High · Evacuation	High · Severe alert	Moderate · Monitor	Low · Baseline safe

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DFRS · PERSONALIZED DECISION SUPPORT

The same storm produces different warnings at different distances

Distance-Based Dynamic Flood Risk Personalization scales risk with proximity to the active river channel, then combines elevation, soil, and land cover.

Sedawatta / Grandpass reach

River distance 45 m · elevation 3.2 m · rapid arrival alert: 3.4 hours · depth 3.84 m.

Malabe elevated reach

River distance 1,250 m · elevation 32 m · low risk score below 25% · normal baseline status.

Decision layer

React dashboard with live SVG risk dials, backend health monitoring, scenario presets, DMC SOP guidance, and Sinhala, Tamil, and English SMS and push alerts.



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CROSS-SYSTEM INTEGRATION · DATA CONTRACT

A twelve-feature hand-off turns Component A's forecast into Component B's decision

COMPONENT A → REST API & IN-PROCESS BRIDGE

Three dynamic hydrometeorological features: rainfall_7d_mm = round(rain_cum72h × 1.8 + rain_24h, 2), cumulative antecedent saturation (200 mm); monthly_rainfall_mm = round(rain_cum72h × 4.2 + 80.0, 2), 30-day baseline ground saturation (350 mm); rainfall_intensity_mmhr = round(iot_rate × 3.5, 2), peak storm cloudburst intensity (15–28 mm/hr).

COMPONENT B → FLOOD MATRIX

Nine geospatial and catchment features: district: Colombo / Gampaha; latitude & longitude: GPS WGS84 coordinates; elevation_m: topographic elevation; distance_to_river_m: proximity to active channel; landcover: Urban, Agriculture; soil_type: Clay, Loamy, Sandy; historical_flood_count: DMC records; river_flow_rate_m3s: upstream hydrometric discharge.

Clicking "Sync IoT Rainfall" instantly binds Component A's forecast via REST API & In-Process Bridge into Component B's Two-Stage XGBoost model.

YES	56.92%	3.41h	3.84m
Occurrence	Risk	Arrival	Height

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EMPIRICAL VALIDATION · COMMUNITY FIELD TRIALS

Three weeks in the field turned user feedback into the next firmware and risk-model iterations



Trial protocol

Sedawatta, Hanwella, and Kaduwela · 15 households, 5 farmers, 2 disaster-response coordinators · 12 rainfall pulses from 12 to 145 mm/day.



User evaluation

4.4 / 5.0 overall satisfaction · 4.6 timeliness · 4.8 language clarity · 4.3 UI usability · false alarms <8.3%.



Post-test iterations

Median filtering reduced wind-blown debris triggers; configurable distance weighting improved elevated-farmer sensitivity; deep-sleep cycling extended battery life to 90 days

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